Large-Scale Data Analysis

Project 2025-2026

1 Introduction

This project addresses a large-scale data classification problem: distinguishing between Higgs boson signal processes and background processes. The primary objective is to evaluate the trade-offs between traditional SciKit-Learn and distributed PySpark pipelines. To ensure a rigorous comparison, we will implement three consistent classification methods across both frameworks, resulting in six tuned models. Our analysis will focus on two key dimensions: predictive performance (Accuracy and ROC scores) and computational efficiency (training and inference time). Through this comparison, we aim to gain insights into the most effective learning strategies for large-scale datasets.

2 Dataset

The Higgs Boson Dataset [1] has been produced using Monte Carlo simulations. The first 21 features (columns 2-22) are kinematic properties measured by the particle detectors in the accelerator. The last 7 features are functions of the first 21 features; these are high-level features derived by physicists to help discriminate between the two classes. There is an interest in using machine learning methods to obviate the need for physicists to manually develop such features.

The standard benchmark results using Bayesian Decision Trees from a standard physics package and 5-layer neural networks are presented in the original paper [2]. The full dataset contains 11,000,000 examples and the last 500,000 examples are used as a test set. In this project, we use an extract of 2,000,000 examples of the full dataset as the training set and 500,000 examples as the test set. These two datasets are available at:

https://utbox.univ-tours.fr/s/dXYJaT2x7fkaxGm

3 Project Tasks

1. Dataset Analysis
    a. Perform a comprehensive analysis of the dataset based on the provided physical descriptions.
    b. Conduct statistical analysis to understand the distribution of signal vs. background processes and identify potential challenges in the raw data.
2. Comparative Feature Engineering Pipelines
    a. Construct two distinct feature engineering pipelines—one using SciKit-Learn and one using PySpark ML.

   b. Both pipelines must implement identical transformations, including feature vectorization (e.g., `VectorAssembler` in Spark) and feature scaling (e.g., `StandardScaler`).
   c. Explain the selection of features and justify why certain attributes are critical for distinguishing Higgs boson production.

3. Model Development, Tuning, and Cross-Validation
   a. Implement three consistent classification methods (e.g., Logistic Regression, Random Forest, and Gradient Boosted Trees, but not limited to) across both frameworks, resulting in a total of six models.
   b. For each method, perform K-fold cross-validation and parameter tuning to identify the optimal model configuration.
   c. Ensure that the search space for hyperparameters is consistent between the SciKit-Learn and PySpark implementations to allow for a fair comparison.
4. Computational Efficiency and Scalability Benchmarking
   a. Conduct a benchmarking study by measuring and reporting the training and prediction times relative to the number of CPU cores utilized.
   b. For SciKit-Learn, control parallelism using the `n_jobs` parameter.
   c. For PySpark, configure the `SparkSession` using different `local[k]` master configurations (e.g., `local[1]` vs. `local[2]`).
   d. Discuss the impact of parallel speedup when handling 2,000,000 training records.
5. Performance Evaluation and Framework Comparison
   a. Apply the six tuned models to the test dataset (500,000 records) and compare their performance using Accuracy and ROC-AUC scores.
   b. Provide a critical analysis of the results, focusing on:
      i. The consistency of predictive accuracy between local and distributed frameworks.
      ii. The time-efficiency trade-offs of using PySpark for large-scale learning vs. SciKit-Learn in a resource-constrained environment like Google Colab.

4 Policy on the Use of Artificial Intelligence (AI)

In this project, AI tools (such as ChatGPT, Claude, Gemini, or GitHub Copilot) are categorized as supportive technologies. While they can enhance productivity, their use must be transparent and limited to technical assistance.

1. Permitted Use of AI

   Students are encouraged to use AI tools for the following purposes:

- Resolving environment configuration issues (e.g., PySpark JVM errors), code debugging or troubleshooting.
- Improving the grammatical flow and professional tone of the final report.

2. Prohibited Use of AI

   The following practices are considered academic misconduct:

   - Generating the entire project notebook or the analytical discussion via AI prompts.
   - Using AI to simulate performance metrics (Accuracy, ROC, Time) or to generate fake experimental plots. All reported data must be derived from actual executions in the Google Colab environment.
   - Submitting AI-generated explanations without verifying their technical accuracy or relevance to this Higgs Boson Dataset.

3. Required Disclosure Statement

   Every submission must include an "AI Usage Appendix" at the end of the report. This appendix should detail:

   - Which AI tools were used (if any).
   - Specific tasks where AI provided any assistance (e.g., "AI was used to debug the `VectorAssembler` error").
   - A certification that all experimental results and comparative analyses are the original work of the team members.

5 Important Notes

The project must be submitted to CELENE as a single ZIP file before 23:00 March 10 2026, including the source code, the trained models, and a report in PDF. The report must be at most 10 pages with unlimited appendix.

References

[1] https://archive.ics.uci.edu/dataset/280/higgs

[2] https://arxiv.org/pdf/1402.4735